



US012258348B2

(12) **United States Patent**
Berecz et al.(10) **Patent No.:** **US 12,258,348 B2**(45) **Date of Patent:** **Mar. 25, 2025**(54) **METHOD FOR THE PRODUCTION OF LUMATEPERONE AND ITS SALTS****309/30** (2013.01); **C07C 309/35** (2013.01);
C07B 2200/07 (2013.01); **C07B 2200/13**
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Budapest (HU)(58) **Field of Classification Search**

None

See application file for complete search history.

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Maier et al: "Separation of Enantiomers: Needs, Challenges, Perspectives", Journal of Chromatography A, vol. 906, Issues 1-2, pp. 3-33, Jan. 2001 ; DOI:10.1016/S0021-9673(00)00532-X.

Primary Examiner — Zinna Northington Davis(74) *Attorney, Agent, or Firm* — Millen, White, Zelano & Branigan, P.C.; Csaba Henter(57) **ABSTRACT**

Method for the production of formula (I) lumateperone or its acid addition salts so that the enantiomer compound with stereochemistry 6bR,10aS

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 92 days.

(21) Appl. No.: **18/110,570**(22) Filed: **Feb. 16, 2023**(65) **Prior Publication Data**

US 2023/0331727 A1 Oct. 19, 2023

Related U.S. Application Data

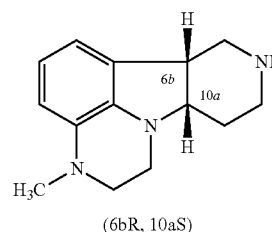
(62) Division of application No. 16/767,057, filed as application No. PCT/HU2018/050050 on Nov. 27, 2018, now Pat. No. 11,655,251.

(30) **Foreign Application Priority Data**

Nov. 27, 2017	(HU)		P1700491
Nov. 27, 2017	(HU)		P1700492
Nov. 27, 2017	(HU)		P1700493
Apr. 16, 2018	(HU)		P1800126

(51) **Int. Cl.****C07D 471/16** (2006.01)**C07B 57/00** (2006.01)**C07C 235/74** (2006.01)**C07C 309/30** (2006.01)**C07C 309/35** (2006.01)(52) **U.S. Cl.**CPC **C07D 471/16** (2013.01); **C07B 57/00**
(2013.01); **C07C 235/74** (2013.01); **C07C**

(II)



is separated from the cis racemate using resolution and the formula (II) stereoisomer is alkylated with 4-halo-4'-fluoro butyrophenone (X=I, Br, Cl) to produce the formula (I) lumateperone, or optionally its acid addition salt. The object of the invention also relates to the amorphous form of the morphologically uniform p-toluenesulfonic acid salt of lumateperone and to the naphthalene-2-sulfonic acid salt of lumateperone, to the 1:2 stoichiometry salt of lumateperone formed with naphthalene-2-sulfonic acid.

20 Claims, 13 Drawing Sheets

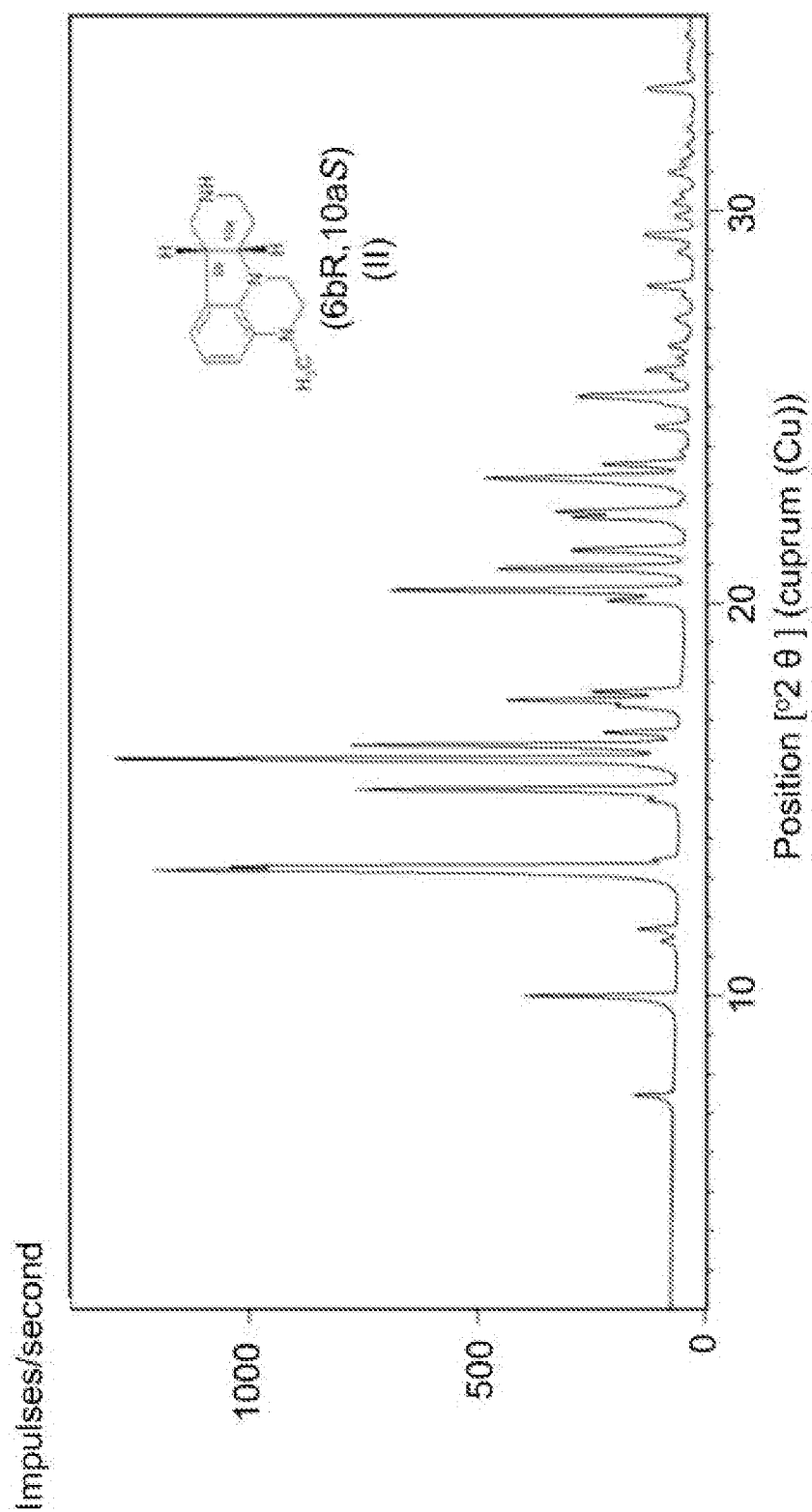


FIG. 1

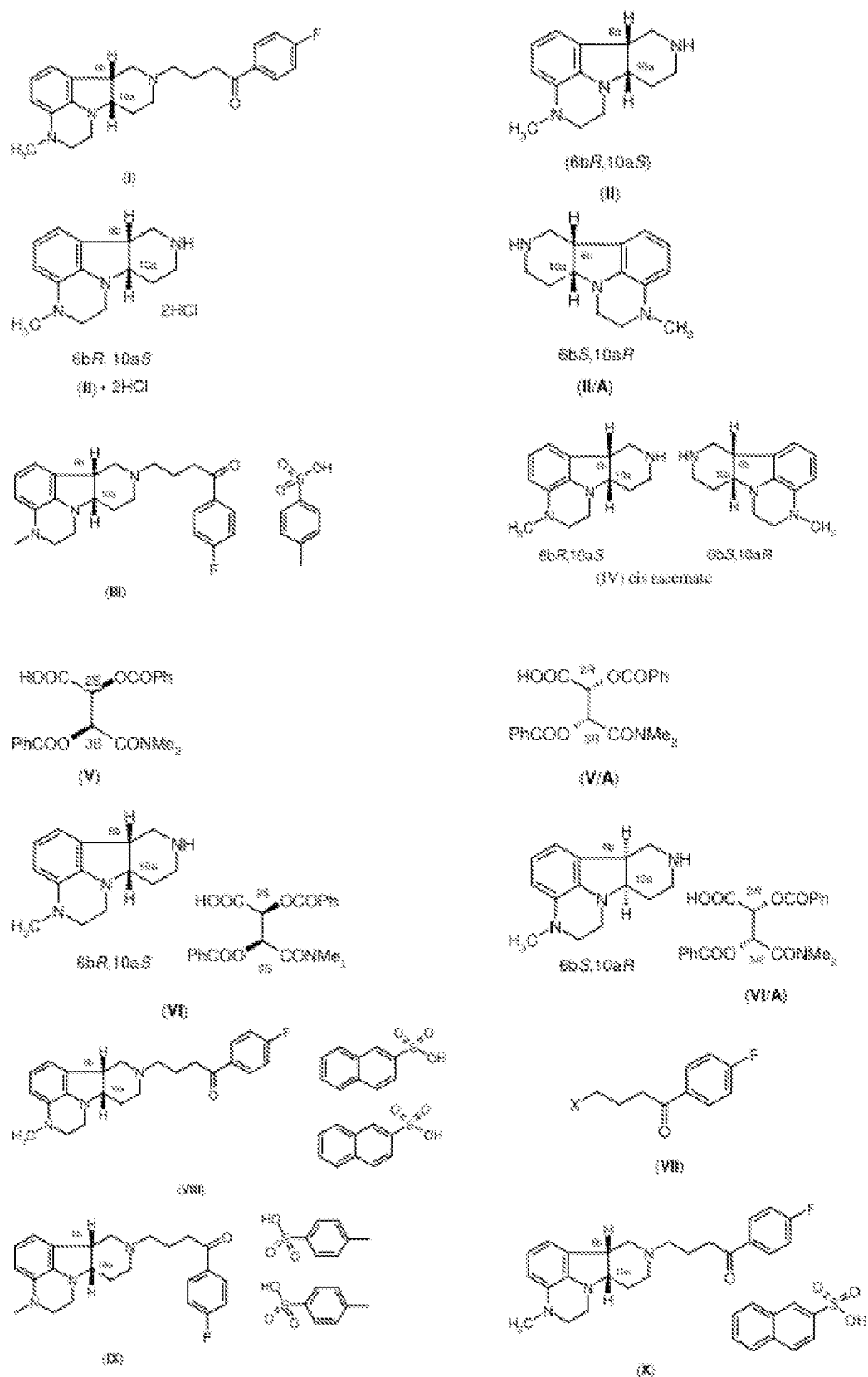


Fig. 9

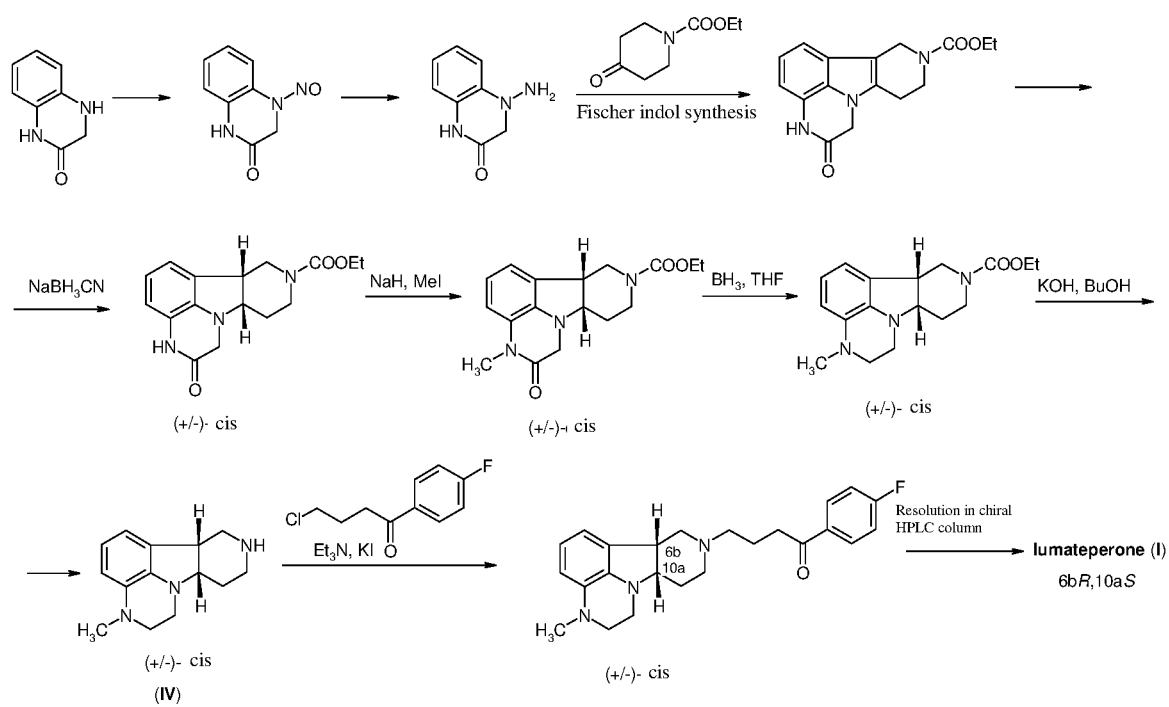


Fig. 11

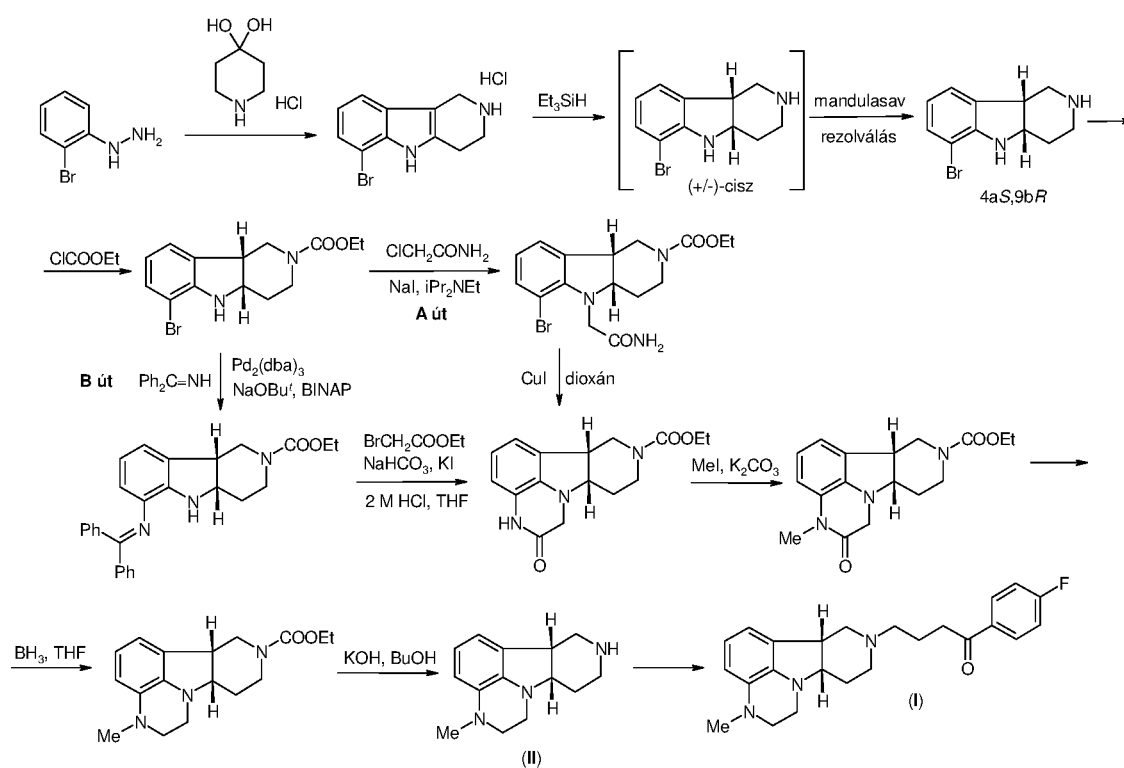


Fig. 12

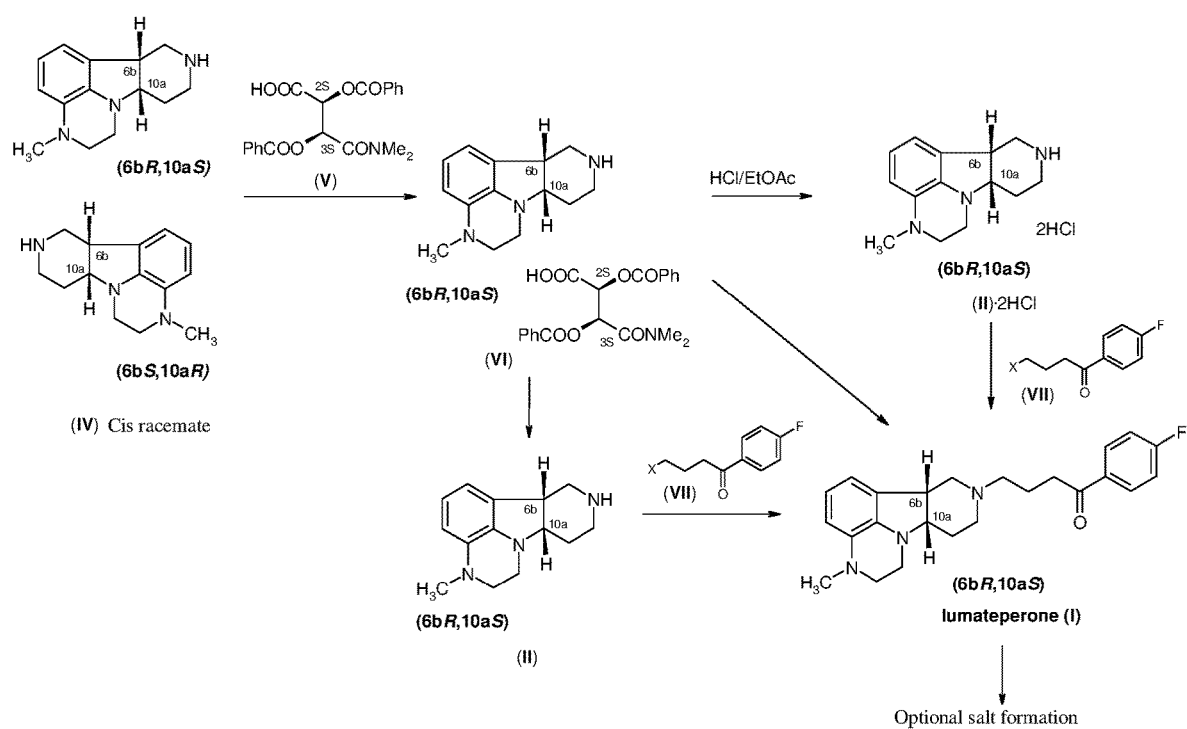


Fig. 13

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did not even reach chemical purity of 97%. In addition, where the yield was given for the salt or cocrystal formation, it was low.

In the patent application number WO 2018/031535 published on 15 Feb. 2018, the authors describe new bis-tosylates. According to page 16 of the application, the lumateperone base or monotosylate salt used for the production of the bis-tosylates may be produced according to that described in patent application number WO 2008/112280, or WO2009/114181. As these processes do not make it possible to make chemically pure product without making use of chromatography methods, the salts presented here could only be produced by using a chiral chromatography technique according to the state of the art. At the same time, although ten salt-production examples are presented in the application, not even one of the salts is characterised using analytical methods apart from x-ray powder diffractograms, and their yields are not even provided. The enantiomer purity and chemical purity are not known about the products obtained either. Therefore, a person skilled in the art is unable to decide whether these salts have any advantage over the previously disclosed monotosylate salts. The state of the art is summarised in the following table:

TABLE 1

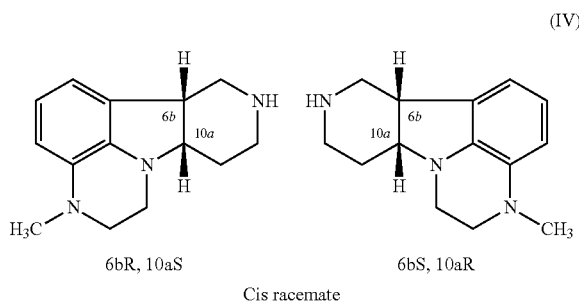
Salt	yield	purity [HPLC]	Literature
Tosylate (polymorph A)	84%	93.20%	WO
Tosylate (polymorph B)	65.72%	96.40%	2009/114181
Tosylate (polymorph B)	81.59%	n.a.	
Tosylate (polymorphology not known)	78.38%	n.a.	
Tosylate (polymorphology not known)	61%	96.90%	
Oxalate salt	n.a.	91%	WO
4-aminosalicylate salt	n.a.	78%	2017/172784
Cyclamate salt	n.a.	67%	
Cocrystal formed with isonicotinamide	n.a.	95%	WO 2017/172811
Cocrystal formed with nicotinamide	n.a.	n.a.	

At present, according to the state of the art, lumateperone suitable for use in the pharmaceutical industry cannot be produced without the use of a chiral chromatography process. In addition, lumateperone salt with chemical purity of 97% cannot be produced either using salt formation. This is the reason why it is necessary to develop an active substance, preferably a lumateperone salt that can be produced with high purity, that is stable and preferably morphologically uniform. The objective of the invention, then, was to overcome the disadvantages of the known methods, especially the chiral purification step and develop a novel method that can be implemented simply and economically at industrial scales with which lumateperone salts can be produced that are suitable for use in pharmaceutical preparations and for the treatment of the aforementioned illnesses/disorders.

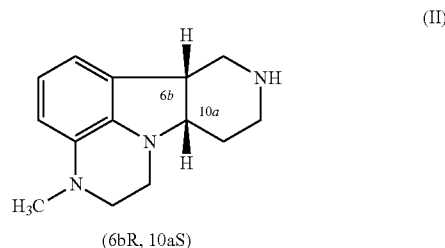
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THE ESSENCE OF THE INVENTION

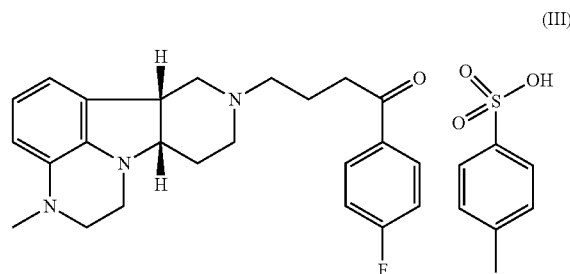
The essence of the invention is a novel method for the production of the formula (I) lumateperone or its acid addition salts so that the enantiomers forming the formula (IV) cis racemate are isolated



and the formula (II) compound obtained in this way with stereochemistry 6bR,10aS

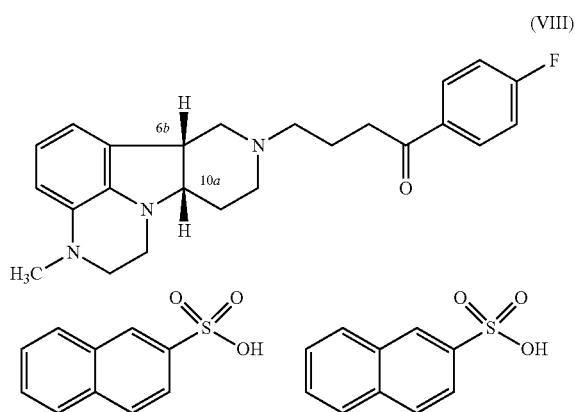


is alkylated with 4-halo-4'-fluoro butyrophenone (X=I, Br, Cl) into formula (I) lumateperone, and optionally transformed into an acid addition salt. According to a preferable embodiment of the invention the lumateperone base is transformed into a salt formed preferably with p-toluenesulfonic acid or with naphthalene-2-sulfonic acid, even more preferably into an amorphous or crystalline salt formed with formula

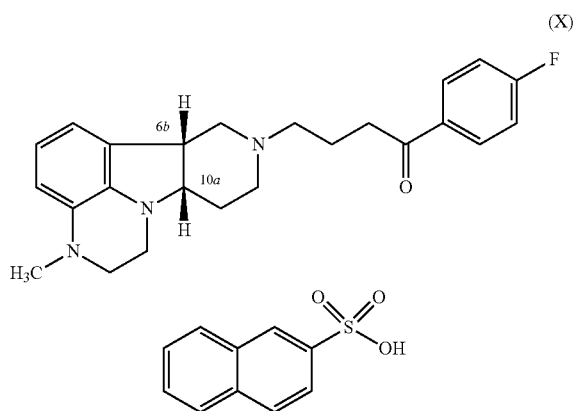


p-toluenesulfonic acid, in which the molar ratio of lumateperone to p-toluenesulfonic acid is 1:1, or into a salt of lumateperone formed with naphthalene-2-sulfonic acid, preferably into the formula (VIII) salt formed with naphthalene-2-sulfonic acid,

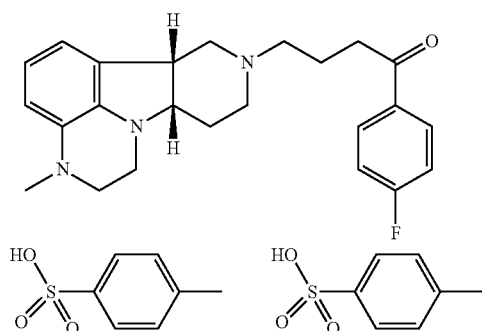
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in which the molar ratio of lumateperone to naphthalene-2-sulfonic acid is 1:2, or into the formula (X) salt formed with naphthalene-2-sulfonic acid,



in which the molar ratio of lumateperone to naphthalene-2-sulfonic acid is 1:1, or into the formula (IX) salt,



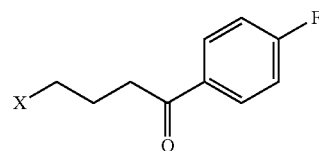
in which the molar ratio of lumateperone to p-toluenesulfonic acid is 1:2. It was surprising to experience that the salts isolated in this way have a chemical purity and stereochemical purity that are suitable for use in the pharmaceutical industry, with a morphological uniform crystalline or amorphous structure and suitable chemical stability.

More specifically, according to a preferable embodiment of the method according to the invention the isolation of the

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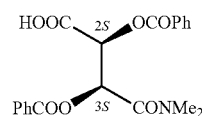
stereoisomers of the formula (IV) compound is performed without the use of a chiral chromatography process.

According to a preferable embodiment of the invention the isolation of the stereoisomers of the formula (IV) compound is performed using a resolution process. The resolution is preferably performed so that the formula (IV) compound is reacted with one of the enantiomers of a chiral acid, then the one stereoisomer salt is isolated from the reaction mixture, and then the formula (II) compound with stereochemistry 6bR,10aS or its salt is alkylated with formula (VII)

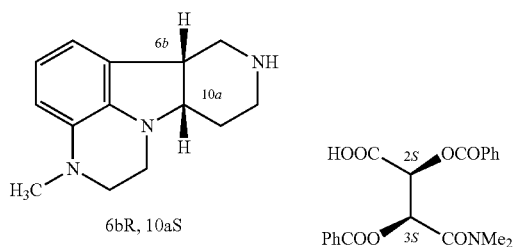


4-halo-4'-fluoro butyrophenone (X=I, Br, Cl) into formula (I) lumateperone, or, optionally, transformed into its acid addition salt. The alkylation step may also be carried out by measuring the formula (II) compound with the stereochemistry 6bR,10aS formed during the resolution step as a base into the reaction mixture used for alkylating, it may also be carried out by measuring it in as a salt, in the presence of an acid-binder and transforming it into a base in the reaction mixture, or by first transforming the salt into a base, then alkylating the base after isolating it from the salt-forming component.

According to a very preferable embodiment of the invention the production of the formula (I) lumateperone or of its acid addition salt is carried out by reacting the formula (IV) compound, which is a cis racemate according to its spatial structure, with the formula (V) compound



then the formula (VI) salt,

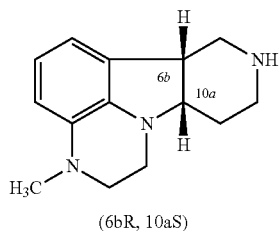


which is the desired stereochemical cis enantiomer salt, is isolated. This enantiomer salt is transformed into a base, or into another salt, such as hydrochloride salt. The salt or base obtained in this way is alkylated. The process may also involve transforming another salt, e.g. made using hydro-

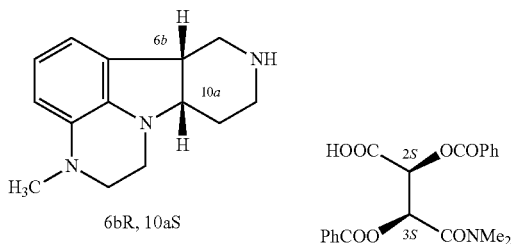
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The invention claimed is:

1. A



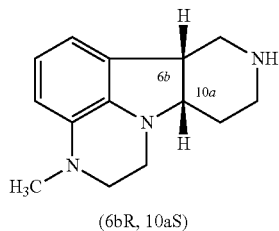
salt of formula (VI)



(6bR, 10aS)-3-methyl-2,3,6b,9,10,10a-hexahydro-1H-pyrido[3',4': 4,5]pyrrolo[1,2,3-de]quinoxaline formed with (+)-dibenzoyl-D-tartaric acid mono(dimethylamide),

or

a salt of formula (II)

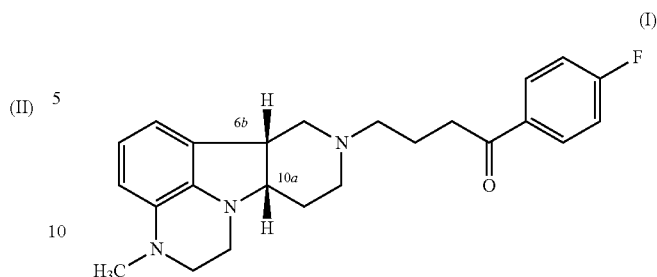


(6bR, 10aS)-3-methyl-2,3,6b,9,10,10a-hexahydro-1H-pyrido[3',4': 4,5]pyrrolo[1,2,3-de]quinoxaline formed with hydrochloric acid,

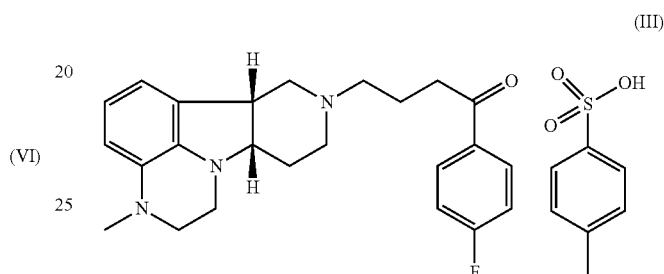
or

the salt of formula (I) lumateperone formed with naphthalene-2-sulfonic acid,

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or
the amorphous form of formula (III) lumateperone salt



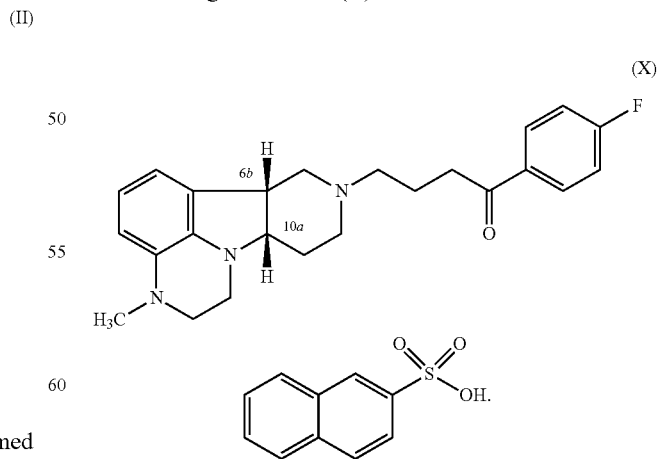
in which salt the molar ratio of lumateperone and the p-toluenesulfonic acid is 1:1.

2. A compound according to claim 1, wherein the enantiomer purity of a compound of formula (II) or (I) or a salt thereof, in which the configuration of the chiral carbon atoms is 6bR,10aS, is greater than 97%.

3. A compound according to claim 1, wherein the chemical purity of a compound of formula (II) or (I) or a salt thereof, in which the configuration of the chiral carbon atoms is 6bR,10aS, is greater than 97%.

4. A compound according to claim 1, which is a salt of formula (I) lumateperone formed with naphthalene-2-sulfonic acid.

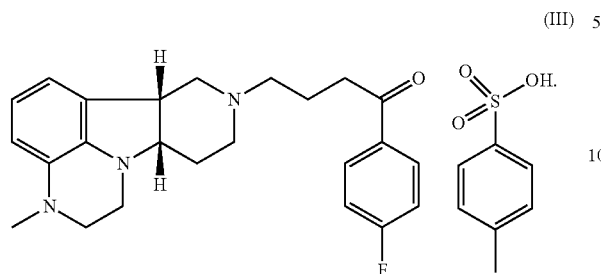
5. A compound according to claim 1, which is a salt of formula (I) lumateperone formed with naphthalene-2-sulfonic acid being of formula (X)



6. A compound according to claim 5, wherein the molar ratio of the lumateperone and the naphthalene-2-sulfonic acid is 1:1.

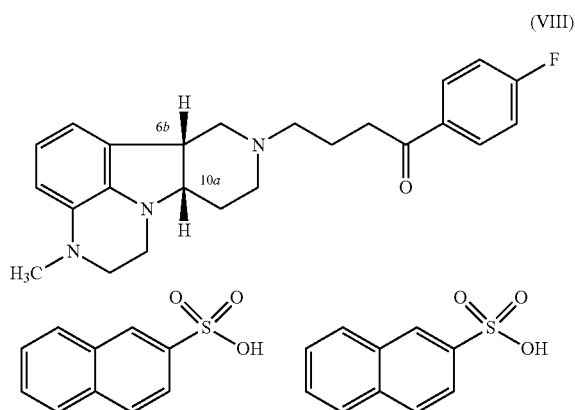
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7. A compound according to claim 1, which is the amorphous form of formula (III) lumateperone salt



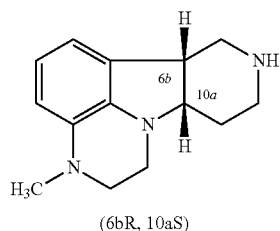
8. A compound according to claim 7, wherein the molar ratio of lumateperone and the p-toluenesulfonic acid is 1:1.

9. A compound according to claim 1, which is a salt of formula (I) lumateperone formed with naphthalene-2-sulfonic acid being of formula (VIII)



in which salt the molar ratio of the lumateperone and the naphthalene-2-sulfonic acid is 1:2.

10. A compound, which is a crystalline form of formula (II)



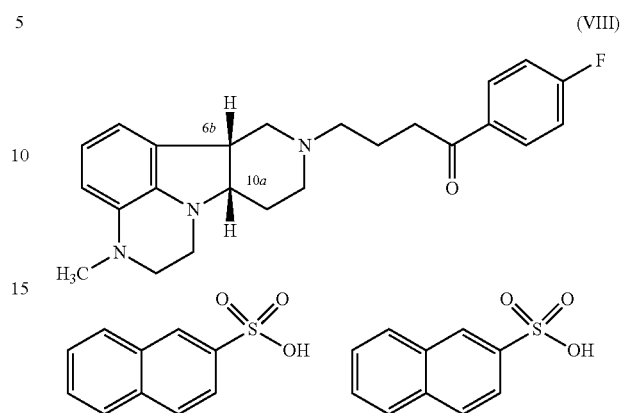
and which is characterized by x-ray powder diffraction lines as follows: [(Cu K α) $^{\circ}2\theta(\pm 0.2^{\circ}2\theta)$]: 9.99; 16.03; 20.87.

11. A compound according to claim 1, which is a salt of formula (VI) and which is characterized by x-ray powder diffraction lines as follows: [(Cu K α) $^{\circ}2\theta(\pm 0.2^{\circ}2\theta)$]: 6.17; 19.88; 23.58.

12. A compound according to claim 1, which is a salt of formula (II) formed with hydrochloric acid and which is characterized by x-ray powder diffraction lines as follows: [(Cu K α) $^{\circ}2\theta(\pm 0.2^{\circ}2\theta)$]: 13.43; 16.52; 22.1.

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13. A compound according to claim 1, which is of formula (VIII)



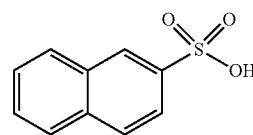
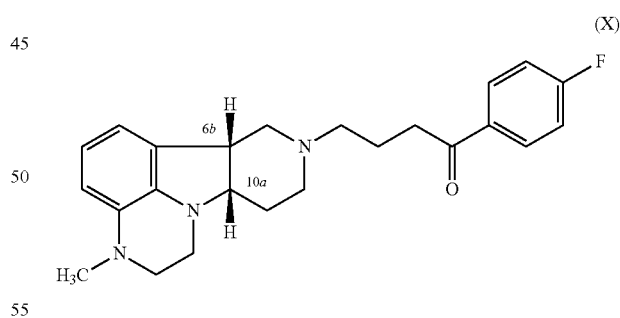
and which is characterized by x-ray powder diffraction lines as follows: [(Cu K α) $^{\circ}2\theta(\pm 0.2^{\circ}2\theta)$]: 3.87; 15.09; 20.71.

14. A compound according to claim 1, wherein the enantiomer purity of a compound of formula (II) or (I) or a salt thereof, in which the configuration of the chiral carbon atoms is 6bR,10aS, is greater than 99.5%.

15. A compound according to claim 1, wherein the chemical purity of a compound of formula (II) or (I) a salt thereof, in which the configuration of the chiral carbon atoms is 6bR,10aS, is greater than 99.5%.

16. A pharmaceutical composition, comprising a compound according to claim 1, which is a salt of formula (I) lumateperone formed with naphthalene-2-sulfonic acid, and a pharmaceutically acceptable excipient.

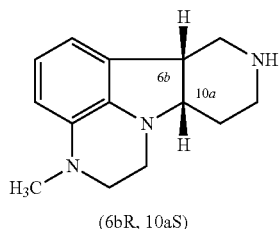
17. A pharmaceutical composition, comprising a compound according to claim 1, which is a salt of formula (I) lumateperone formed with naphthalene-2-sulfonic acid being of formula (X)



or which is the amorphous form of formula (III) lumateperone salt, and a pharmaceutically acceptable excipient.

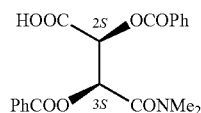
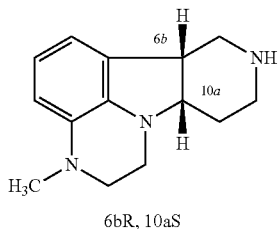
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18. A method for producing the crystalline form of formula (II)



(6bR, 10aS)-3-methyl-2,3,6b,9,10,10a-hexahydro-1H-pyrido[3',4':4,5]pyrrolo[1,2,3-de]quinoxaline base, comprising

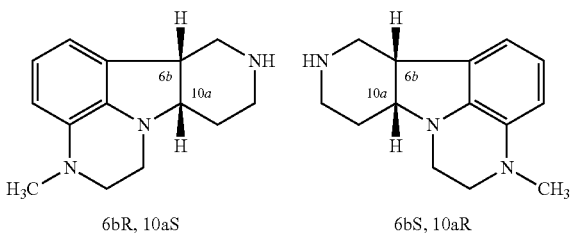
a.) reacting the compound of formula (VI) salt



formed with (+)-dibenzoyl-D-tartaric acid mono(dimethylamide), or the compound of formula (II) salt formed with hydrochloric acid, with a base, and the resultant released formula (II) compound is brought into crystalline form,

or

b.) reacting the formula (IV) cis racemate compound



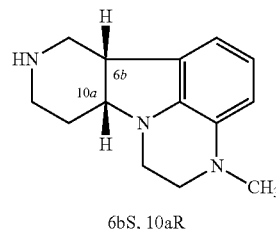
with a chiral acid, then a salt of formula (II) compound formed with a chiral acid, in which the configuration of the chiral carbon atoms is 6bR,10aS, is isolated from a reaction mixture and then transformed into formula (II) base,

or

c.) reacting the formula (IV) compound with chiral acid, then a salt of formula (II/A) compound formed with a chiral acid, wherein the configuration of the chiral carbon atoms is 6bS,10aR,

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(II/A)

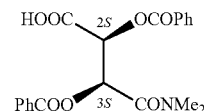


is separated from a reaction mixture, and the resultant formula (II) compound that has been enriched in the reaction mixture, wherein the configuration of the chiral carbon atoms is 6bR,10aS, is isolated, then the formula (II) compound obtained is crystallized;

or

for producing the formula (VI) salt of (6bR,10aS)-3-methyl-2,3,6b,9,10,10a-hexahydro-1H-pyrido[3',4':4,5]pyrrolo[1,2,3-de]quinoxaline formed with (+)-dibenzoyl-D-tartaric acid mono(dimethylamide), comprising

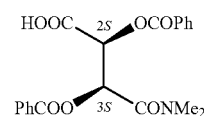
a.) reacting the formula (IV) compound with the formula (V) (+)-dibenzoyl-D-tartaric acid mono(dimethylamide),



then the formula (VI) compound obtained is isolated from a reaction mixture,

or

b.) reacting the formula (IV) compound with a chiral acid, then the salt of compound (II/A) formed with chiral acid, in which the configuration of the chiral carbon atoms is 6bS,10aR, is isolated from a reaction mixture, and the resultant formula (II) compound enriched in the reaction mixture is reacted with the formula (V) (+)-dibenzoyl-D-tartaric acid mono(dimethylamide),



then the formula (VI) compound formed is isolated;

or

for producing a formula (II)·2HCl salt, comprising stirring the formula (VI) salt of (6bR,10aS)-3-methyl-2,3,6b,9,10,10a-hexahydro-1H-pyrido[3',4':4,5]pyrrolo[1,2,3-de]quinoxaline formed with (+)-dibenzoyl-D-tartaric acid mono(dimethylamide) or the formula (II) base in an organic solvent to form a reaction mixture, then hydrochloric acid dissolved in an organic solvent is added to the reaction mixture, then the formula (II)·2HCl salt formed is separated.

19. A method for preparing a salt of lumateperone formed with p-toluenesulfonic acid according to claim 1, compris-